



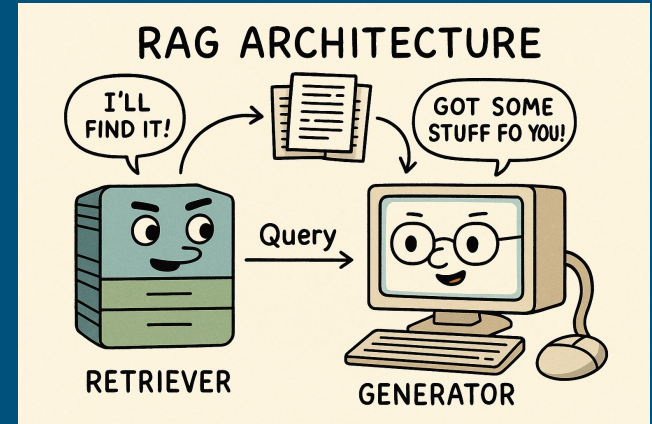
RAGWorks

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Introduction: Our Project

- Our project uses RAG at its core
- RAG in a nutshell
 - RAG = Retrieval-Augmented Generation
 - Retrieve: find useful information
 - Augment: provide the LLM with the retrieved information
 - Generate: LLM will use the knowledge to create a context-aware response to the user
- Our system allows users to:
 - Upload documents that the LLM can use to generate answers
 - Automatically preprocess and embed documents into a vector database
 - Ask questions in natural language & get grounded, source-referenced answers



Main Project Components

Our RAG system is split into 3 main components:

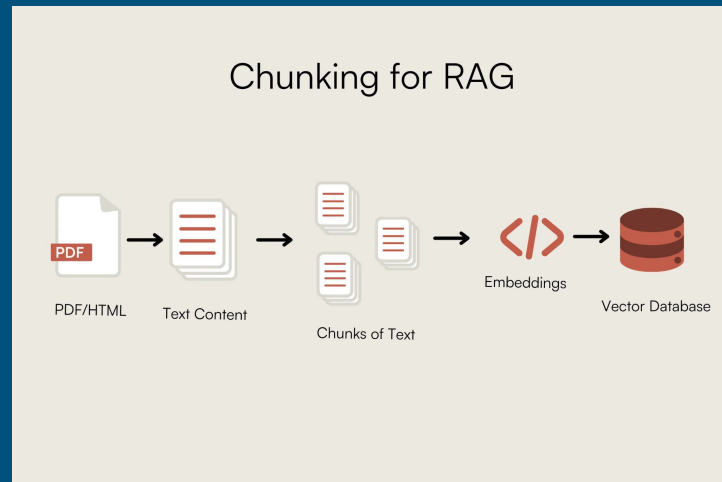
- Document Ingestion
 - Upload documents, document is split into chunks
- Embedding and Vector Storage
 - Store document chunks in the vector database (ChromaDB) where embeddings are generated
- Retrieval/Generation
 - RAG system searches the vector database for embeddings/vectors that are relevant to the user query and passes the context to the LLM
 - The LLM uses the context to generate a response

Why use RAG? Who benefits from this?

- You can control the information that is provided to the LLM to generate responses via documents
- The answers are source-referenced, so you know exactly where the information is coming from
- Who benefits most from RAG:
 - Anyone who wants customized information retrieval based on their own documents. Examples:
 - Developers and organizations wanting domain-specific RAG systems
 - Researchers and educators looking for custom Q&A systems
 - Teams needing structured, document-based AI assistants

Document Ingestion

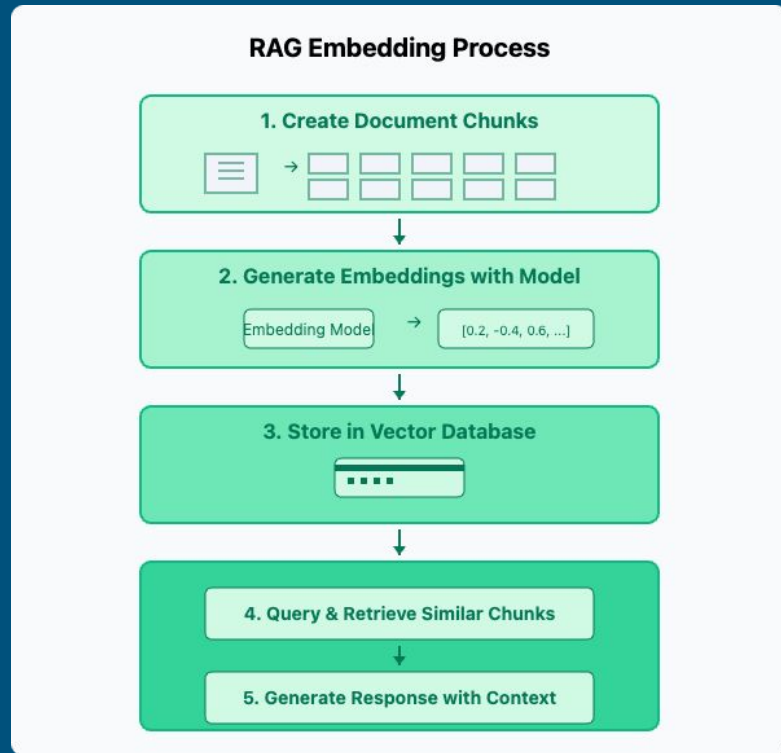
- Accepts various document types (HTML, DOCX, text, pdf, and PPTX)
- Flow of document ingestion:
 - Admin uploads a document
 - If the document has already previously been uploaded, admin will be alerted. There is an option to replace the file if it already exists. Otherwise, if the file has not yet been uploaded, the file path and document type will be verified
 - Once verified, all types of documents are parsed into plain text



- Document is split into chunks that are 800 characters with an overlap of 150 characters
- Example: Chunk 1: 0-799, Chunk 2: 650-1449
- Each chunk is embedded/converted into a vector and stored in ChromaDB

Embedding and Vector Storage

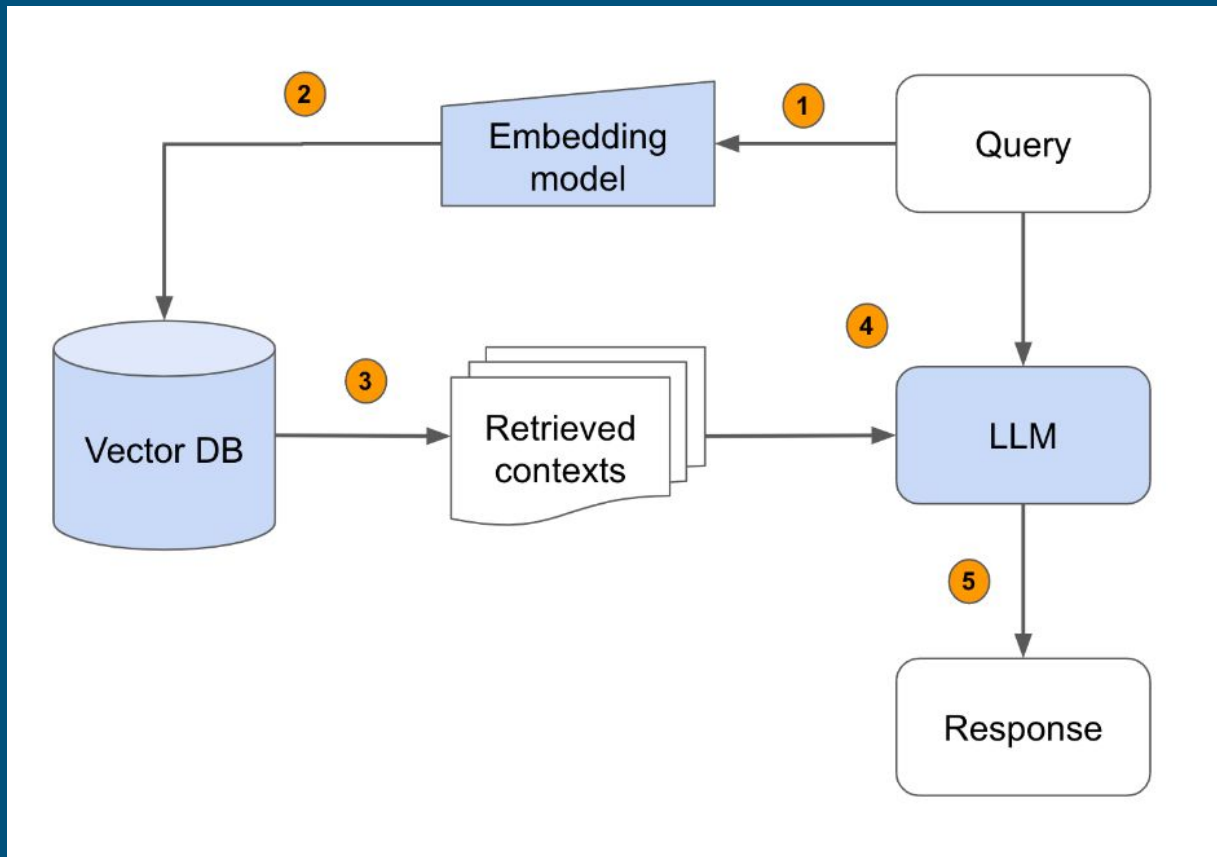
- Embeddings = document chunks converted into vectors
- Our system uses ChromaDB as the persistent vector database
- MySQL Server database is used to unify content and share embeddings amongst users on different local machines
- Embeddings are generated using OpenAI



Query Handling and Answer Generation

- When there is a user query, the user query is converted into an embedding (vector) and ChromaDB searches its database using vector similarity search to find the top k chunks (the most relevant chunks for the query)
- The retrieved chunks are bundled together into a context string that is passed to the LLM
- OpenAI (our LLM) generates an answer from the context
- We have provided the LLM with specific rules for handling the user query:
 - “Answer the question using ONLY the provided context. If the answer is not in the context, say you do not know. When possible, mention the source used.”

Flow Query Handling and Answer Generation



Example User Query and Answer Generation

Query with available context:

AI Chat

How do I graduate from Metropolitan State University?

Send

Response:

To graduate from Metropolitan State University, you must qualify based on the definitions provided in University Policy 2020 and meet the following criteria: 1. Complete a minimum of 30 upper division credits in graded (A-F courses) at Metropolitan State University. 2. Maintain a cumulative GPA that meets the honors designation thresholds if applicable: - Summa Cum Laude: 3.900 and above - Magna Cum Laude: 3.800 to 3.899 - Cum Laude: 3.700 to 3.799 3. Your cumulative GPA will be calculated based on all graded courses taken at the university. 4. You must meet the graduation standards by the end of the semester preceding the semester in which you intend to graduate. (Source: 2520_GradwithHonors.pdf)

Query without available context:

AI Chat

What can you tell me about the University of Minnesota?

Send

Response:

I do not know.

Evaluation of RAG System

- What are the metrics used by the system?
 - **Top K? How many chunks are retrieved per query**
 - Low K = faster & less noise
 - High K = slower & more noise
 - **Temperature? Grounded in logic or creative.**
 - Low temperature = less creative
 - High temperature = more creative
 - **Top P? Freedom of vocabulary**
 - Values 0-1.
 - 1.0 considers all possible word options
 - 0.7 only considers top 70% similar words

```
def run_rag(user_input):  
    result = build_rag_response(user_input, top_k=3, temperature=0.0, top_p=1.0)  
    return result["answer"]
```

How did we test?

- RAG'd 20 Metro State documents
- 30 questions related to documents
- Target answer, target source
- Values on 0 & 1
- LLM Judge for semantic test based on logic and intent

Admin Dashboard Logout

CHROMADB STATS

COLLECTION: RAG_Work_Test UNIQUE DOCUMENTS: 20 TOTAL CHUNKS: 219

UPLOAD DOCUMENT

Enter full file path (e.g., C:\...\File.pdf) pdf ☐ Replace if exists Upload

DOCUMENTS IN CHROMADB Refresh

NAME	UPLOADED	SOURCE PATH	
Find_Advisor.pdf	4/10/26, 4:57:37 PM	[REDACTED]	Delete Update
Waitlist_and_CourseCancellations.pdf	3/17/2026, 8:30:12 PM	[REDACTED]	Delete Update
Transfer_Student_Admissions.pdf	3/17/2026, 8:29:56 PM	[REDACTED]	Delete Update
Transcript_Request_Form.pdf	3/17/2026, 8:29:38 PM	[REDACTED]	Delete Update
Summer2026_AcademicCalendar.pdf	3/17/2026, 8:29:18 PM	[REDACTED]	Delete Update
StudentLife_and_Leadership_Development.pdf	3/17/2026, 8:28:58 PM	[REDACTED]	Delete Update

```
# --- Academic Calendar ---
{
  "question": "When does the Summer 2026 semester begin?",
  "target_answer": "The Summer 2026 semester begins around mid-May, with sessions such as the first session starting around May 16.",
  "target_source": "Summer2026_AcademicCalendar.pdf"
},

# --- Transcript ---
{
  "question": "How can a student request a transcript?",
  "target_answer": "Students can request a transcript through methods such as mail, in-person requests, or through student services, often",
  "target_source": "Transcript_Request_Form.pdf"
},

# --- Question 30 ---
# --- Transfer Admissions ---
{
  "question": "What are the requirements for transfer student admission?",
  "target_answer": "Transfer student admission requires submitting transcripts from previous institutions and meeting academic requirements",
  "target_source": "Transfer_Student_Admissions.pdf"
},
```

Results

- Ran 2 different times 15 questions each with both taking around 8 minutes each.
- Separated for peace of mind, could be done at large scale with more questions and data with safe parameters and trustworthy hardware.

Run 1

```
=====
Starting evaluation run 12/12
Running eval: top_k=8, temp=0.2, top_p=1.0
Progress: 100.0%
Elapsed Time: 443.0s
Estimated Remaining Time: 0.0s
=====

All evaluations completed. Results saved to eval_results_grid.json

===== FINAL RESULTS =====
top_k=2, temp=0.0, top_p=0.8 | Answer Acc=0.73 | Source Acc=1.00
top_k=2, temp=0.0, top_p=1.0 | Answer Acc=0.73 | Source Acc=1.00
top_k=2, temp=0.2, top_p=0.8 | Answer Acc=0.73 | Source Acc=1.00
top_k=2, temp=0.2, top_p=1.0 | Answer Acc=0.73 | Source Acc=1.00
top_k=5, temp=0.0, top_p=0.8 | Answer Acc=0.80 | Source Acc=1.00
top_k=5, temp=0.0, top_p=1.0 | Answer Acc=0.80 | Source Acc=1.00
top_k=5, temp=0.2, top_p=0.8 | Answer Acc=0.80 | Source Acc=1.00
top_k=5, temp=0.2, top_p=1.0 | Answer Acc=0.80 | Source Acc=1.00
top_k=8, temp=0.0, top_p=0.8 | Answer Acc=0.80 | Source Acc=1.00
top_k=8, temp=0.0, top_p=1.0 | Answer Acc=0.80 | Source Acc=1.00
top_k=8, temp=0.2, top_p=0.8 | Answer Acc=0.80 | Source Acc=1.00
top_k=8, temp=0.2, top_p=1.0 | Answer Acc=0.80 | Source Acc=1.00
```

Run 2

```
=====
Starting evaluation run 12/12
Running eval: top_k=8, temp=0.2, top_p=1.0
Progress: 100.0%
Elapsed Time: 499.3s
Estimated Remaining Time: 0.0s
=====

All evaluations completed. Results saved to eval_results_grid.json

===== FINAL RESULTS =====
top_k=2, temp=0.0, top_p=0.8 | Answer Acc=0.83 | Source Acc=0.89
top_k=2, temp=0.0, top_p=1.0 | Answer Acc=0.83 | Source Acc=0.89
top_k=2, temp=0.2, top_p=0.8 | Answer Acc=0.83 | Source Acc=0.89
top_k=2, temp=0.2, top_p=1.0 | Answer Acc=0.83 | Source Acc=0.89
top_k=5, temp=0.0, top_p=0.8 | Answer Acc=0.83 | Source Acc=0.94
top_k=5, temp=0.0, top_p=1.0 | Answer Acc=0.83 | Source Acc=0.94
top_k=5, temp=0.2, top_p=0.8 | Answer Acc=0.83 | Source Acc=0.94
top_k=5, temp=0.2, top_p=1.0 | Answer Acc=0.83 | Source Acc=0.94
top_k=8, temp=0.0, top_p=0.8 | Answer Acc=0.94 | Source Acc=1.00
top_k=8, temp=0.0, top_p=1.0 | Answer Acc=0.94 | Source Acc=1.00
top_k=8, temp=0.2, top_p=0.8 | Answer Acc=0.94 | Source Acc=1.00
top_k=8, temp=0.2, top_p=1.0 | Answer Acc=0.89 | Source Acc=1.00
=====
PS C:\Users\Eli\Downloads\Spring_2026_SEM\RBG_Project\RBG-App\backend>
```

Interpretation

Which parameter combination provided most accurate results?

- Top_k = 8
- Temperature = 0.0
- Top_P = 0.8 - 1.

Top K	Answer Accuracy	Source Accuracy
2	0.73–0.83	0.89–1.00
5	0.80–0.83	0.94–1.00
8	0.80–0.94	1.00

Conclusion

Hallucination in Open Source AI (Chat GPT).

- Grounded response from real documents
- Source references provided
- Limiting known context

Getting the most value out of a RAG model.

- You control the data!

